i want you to do deep research on the company cohere,

how did they origin, what was their vision and mission, what was their selling point, how did they become big, who is in their team, how strong are they, what all projects have they done, what is their goal and mission now,
what kind of roles are open now, do they hire new grads, if they do what are they looking for in them, and any other information that i need to know

Cohere is a Toronto-founded AI company (2019) that builds large language models and enterprise AI platforms, started by researchers behind the original Transformer paper to bring powerful language models safely and reliably into real-world businesses. It focuses on being an independent, enterprise-first AI provider with strong privacy, security, and multi-cloud flexibility rather than a consumer chatbot company.[1] [2] [3] [4] [5]

Origin and founding story

Cohere was founded in 2019 in Toronto by Aidan Gomez, Nick Frosst, and Ivan Zhang, all University of Toronto alumni, with Gomez and Frosst previously working as researchers at Google Brain. Gomez was a co-author of the 2017 “Attention Is All You Need” paper that introduced the Transformer architecture, which underpins modern LLMs and Cohere’s own models.[2] [3] [4] [6]

The founders started Cohere to take large language models out of pure research and make them usable in industry, at a time when LLMs were still seen as mostly academic. Early on, the company focused on building the infrastructure and training pipelines to run large models on available supercomputers, before specific products were fully defined.[3] [4] [5]

Early vision, mission, positioning

Cohere’s early vision was to “scale intelligence to serve humanity” by giving technology the ability to understand and generate natural language for practical business use. The company positioned itself as a “model factory” and platform that enterprises could use to access, customize, and safely deploy state-of-the-art language models.[4] [5] [1]

A key selling point from the beginning was being independent of hyperscaler ecosystems, so enterprises could adopt LLMs with strong data privacy, sovereignty, and flexibility across their preferred cloud providers and infrastructure. Another early focus was enterprise-grade reliability and control (not just raw model performance), including fine-tuning, integrations, and governance.[5] [3] [4]

How they became big and how strong they are now

Cohere grew through three main levers: strong research pedigree, large funding rounds, and a clear enterprise-only strategy. Its leadership team added experienced operators such as Martin Kon (former YouTube CFO) as President and COO, signaling a push to scale commercialization and go-to-market.[7] [6] [3] [4]

The company has raised substantial venture funding (hundreds of millions of dollars) from major investors, reflecting its position as one of the leading LLM providers outside Big Tech. Technically, Cohere's Command-series models, including Command R+, and its multilingual models are competitive on benchmarks for retrieval-augmented generation, tool use, and enterprise workloads, which underpins its strength with large customers.[6] [3] [4] [7]

Products, projects, and platforms

Cohere builds and sells several categories of products aimed at enterprise AI:

- Foundation and instruction-tuned language models, including the Command family, optimized for tasks such as chat, summarization, search, and agents, with long context windows and multilingual support.[1] [4]
- Retrieval-augmented generation (RAG) tooling and APIs, enabling enterprises to ground models on their internal data with better factuality and governance.[4]
- Enterprise AI platform offerings (including a newer workplace-productivity platform called North) that provide an end-to-end environment for deploying AI assistants, search, and automation across organizations.[1] [4]

On the research and community side, Cohere launched Cohere Labs in 2022 as an open science initiative, which now has thousands of community members and over 100 research papers published. The company also runs collaborations and integrations with major cloud and software vendors to embed its models into broader enterprise stacks.[5] [4] [1]

Current mission, focus, and differentiation

Cohere's current mission centers on building AI that enhances human judgment, accelerates execution, and extends what individuals and teams can achieve—essentially amplifying knowledge work rather than replacing it. The company explicitly frames its aim as scaling intelligence in ways that are safe, secure, and aligned with enterprise requirements for privacy and control.[3] [4] [1]

Its main differentiation points today include:

- Independence from any single cloud or consumer ecosystem, appealing to enterprises that want neutral AI infrastructure.[3] [4] [5]
- Strong focus on data privacy, security, and compliance, positioning LLMs as safe to use with sensitive corporate and government data.[4] [5] [3]
- Deep expertise in language-centric models and RAG, rather than a sprawling set of unrelated AI products.[4]

Team and leadership

The founding team is: Aidan Gomez (CEO), Nick Frosst, and Ivan Zhang, all with backgrounds in deep learning and NLP. Gomez brings research leadership from Google Brain and academic work at Oxford; Frosst also worked at Google Brain; Zhang previously collaborated with Gomez on earlier startups and research.[8] [2] [3] [4]

Beyond the founders, the executive team includes leaders such as President/COO Martin Kon (ex-YouTube CFO) and other senior leaders spanning product, research, engineering, and go-to-market roles, giving the company a blend of deep technical and business scaling experience. Cohere's broader team includes researchers, engineers, product managers, and sales/solutions staff distributed across North America and Europe, with a strong presence in Toronto.[1] [3] [4]

Hiring, roles, and new-grad opportunities

Cohere currently lists roles across engineering, machine learning, research, product, GTM, and business functions, including titles like Member of Technical Staff (MLE), software engineer, research scientist, and multiple internship positions. Their public jobs feed shows categories such as Engineer, Scientist, Software Engineer, Machine Learning Engineer, Product Designer, Analyst, Manager, and Intern roles.[9]

The company does hire early-career talent through research internships and other intern roles, and some junior or "Member of Technical Staff" postings may be accessible to strong new grads depending on region and team. For new grads, patterns across roles emphasize:[9]

- Strong fundamentals in computer science, math, and machine learning (e.g., deep learning, NLP/LLMs, distributed training).[9] [3] [4]
- Practical experience via research projects, open-source work, or production systems, plus comfort with modern ML tooling and languages (Python, PyTorch/JAX, cloud infra).[3] [9] [4]

What they tend to look for in candidates

From role descriptions and the company's culture, candidates who stand out typically show:

- Demonstrated impact in ML or systems work, such as shipping models, improving training pipelines, or publishing solid research.[9] [3] [4]
- Ability to work on end-to-end problems in small, high-ownership teams, including collaboration with product and customers.[3] [4] [9]
- Interest in enterprise applications rather than just consumer chatbots—understanding use cases like search, knowledge assistants, and internal tooling.[5] [4] [3]

Soft-skill signals include clear communication, comfort with ambiguity in fast-moving research and product areas, and alignment with Cohere's mission of building responsible, secure AI for organizations.[1] [4] [3]

How to prepare if you want to apply

For a new grad or near-new grad targeting Cohere-type roles, steps that align well with what they hire for include:

- Build 1–2 solid portfolio projects around LLMs or RAG (e.g., retrieval-based QA over custom data, evaluation frameworks, or fine-tuning experiments) on GitHub. Ground them in practical use cases that could matter to enterprises.
- Develop strong fundamentals in deep learning and NLP (transformers, attention, optimization, evaluation) and be able to reason about tradeoffs in model size, latency, and cost.
- Practice articulating how you'd bring LLMs into a real organization: data pipelines, privacy, evaluation, safety, and failure modes.

If you share your current degree level, graduation date, and target role (e.g., research vs. product-focused MLE), a tailored prep plan and resume keyword list for Cohere-style roles can be outlined next.

⁂

okay I need more information on what are the responsibilities of all the role that they are open to hire, also find profiles that have got those roles and understand what kind of projects are they expecting to have for each of these roles. give three projects for each role

Here are the **responsibilities and project expectations** for key open roles at Cohere, plus examples of profiles and relevant projects for each:

1. Member of Technical Staff (MLE, Modeling, Data, Efficiency, Multimodal)

Responsibilities:

- Plan and execute major ML projects from ideation to production
- Align cross-disciplinary teams (engineering, product, legal, partners)
- Design, train, and optimize advanced models (LLMs, multimodal, search)
- Identify gaps in existing models, strategize improvements
- Mentor other engineers and researchers
- Collaborate with partners and legal teams on patents/IP
- Work hands-on with distributed training, inference, ML infrastructure[11] [12] [13] [14] [15]

Profile Examples & Typical Projects:

- *Profiles*: Ex-ML engineers from Meta, Google, DeepMind; PhDs in NLP/ML; Open-source contributors on HuggingFace, TensorFlow, PyTorch[14] [15] [11]
- *Projects Expected*:
  - Developed and optimized a retrieval-augmented generation pipeline for internal enterprise search
  - Created efficiency benchmarks and implemented novel model compression techniques for LLMs
  - Built and deployed a multimodal model that handles both text and speech for customer insights

2. Software Engineer / Staff Software Engineer (Infra, Applied ML, Platform)

Responsibilities:

- Build and optimize GPU/Kubernetes infrastructure for large-scale AI training
- Create and maintain the backend of generative AI platforms (Command, North)
- Work on cross-team collaboration for deployments, product lifecycle[16]

Profile Examples & Typical Projects:

- *Profiles*: Systems engineers strong in distributed computing, cloud infra (AWS, GCP, Azure); contributors to ML pipelines in production[16]
- *Projects Expected*:
  - Built HPC orchestration layer for distributed LLM training with GPU resource optimization
  - Developed secure authentication and user management for an enterprise AI platform
  - Designed ML pipeline automation for continuous deployment and monitoring

3. AI Engineer / Applied ML Engineer

Responsibilities:

- Apply Cohere's LLMs to real-world problems (semantic search, summarization, agents)
- Design and implement scalable AI workflows and product integrations
- Collaborate across product, engineering, and enterprise customers[13] [11] [16]

Profile Examples & Typical Projects:

- *Profiles*: ML engineers with experience fine-tuning LLMs, integrating models into SaaS platforms
- *Projects Expected*:
  - Deployed an NLP-powered semantic search API for a financial services customer
  - Built chat summarization features using instruction-tuned LLMs for knowledge workers

- Engineered an agent-based system that automates document review

4. Research Scientist / Senior Research Scientist / Research Scholar

Responsibilities:

- Drive independent, high-impact research on frontier AI/ML topics
- Publish findings openly (papers, OSS), advance scientific and societal dimensions of ML
- Collaborate on advanced model and pretraining designs (multilingual, multimodal, retrieval) [17] [18]

Profile Examples & Typical Projects:

- *Profiles*: Published authors in NeurIPS/ICLR, open-source maintainers, postdoctoral ML researchers[18] [17]
- *Projects Expected*:
  - Published a paper on scalable multilingual representation learning with AI agents
  - Designed and benchmarked retrieval-augmented LLMs for enterprise document QA
  - Released and documented an open-source toolkit for multimodal sentiment analysis

5. Product Designer / Technical Product Manager

Responsibilities:

- Design user experience for enterprise AI platforms (North, Compass)
- Define product strategy and roadmaps for AI-driven solutions
- Conduct user research, engage cross-functionally in feature rollout[19] [16]

Profile Examples & Typical Projects:

- *Profiles*: UX/product designers with prior SaaS experience, technical PMs in B2B ML startups
- *Projects Expected*:
  - Led UI/UX overhaul for LLM-powered internal search product
  - Defined roadmap and delivered MVP for agent-based automation platform
  - Conducted user research and shipped onboarding flows for enterprise AI solutions

6. Solutions Architect / Analyst / Account Manager

Responsibilities:

- Own technical and customer relationships for enterprise AI implementations
- Lead proofs-of-concept (POCs), map customer needs to Cohere's NLP tools

- Bridge technical and client-facing teams, manage onboarding[20]

Profile Examples & Typical Projects:

- *Profiles*: Technical consultants, experienced analysts in ML adoption, client-facing engineers[20]
- *Projects Expected*:
  - Launched a pilot for RAG-based internal knowledge systems at a Fortune 500 firm
  - Designed benchmarking framework for enterprise LLM robustness evaluations
  - Managed onboarding and integration for customer NLP pipelines

Additional Notes:

- Cohere expects candidates, regardless of role, to own impactful projects that involve production deployment, deep collaboration, and adaptation of models to real-world business problems.
- Published research, open-source contributions, and deep ownership of end-to-end ML solutions are highly valued across roles.[11] [14] [17] [18]
- You'll find diverse teams, with backgrounds spanning top tech firms and academia, but strong software and ML fundamentals plus enterprise problem-solving are consistent threads.

If you want details for another specific Cohere role or sample resume keywords for applications, please indicate!

⁂

so summaries to me all the topics i need to know to get a position in this company, using a roadmap

Here's a **roadmap of all the topics and skills you need to know to become a strong candidate for Cohere,** particularly for technical, ML, or engineering roles. This outline covers essentials for new grads and early-career professionals targeting roles such as Member of Technical Staff (MLE), AI Engineer, or Research Scientist.

Cohere Candidate Roadmap: Skills & Topics

1. Core Computer Science & Software Engineering

- Data structures & algorithms (arrays, trees, graphs, searching, sorting)
- Object-oriented programming (OOP) principles
- Version control (Git/GitHub), software debugging, code reviews
- Writing clean, efficient, and well-tested code (Python is a must, others desirable: C++, Go, JS)

2. Machine Learning Fundamentals

- Supervised, unsupervised, and semi-supervised learning basics
- Model evaluation: metrics, train/validation/test splits, cross-validation
- Loss functions, regularization, optimization (SGD, Adam)
- Understanding and applying ML libraries (scikit-learn, TensorFlow, PyTorch)
- Familiarity with cloud ML pipelines (e.g., AWS Sagemaker, GCP Vertex, Azure ML)

3. Deep Learning & Language Modeling

- Neural network theory: MLPs, CNNs, RNNs, attention mechanisms
- Transformers: core architecture, self-attention, positional encoding
- Pretraining, fine-tuning and prompt engineering
- Large language models (LLMs): tokenization, embeddings, context windows, instruction tuning
- Retrieval-augmented generation (RAG): how to ground models on external data

4. Natural Language Processing (NLP)

- Text preprocessing: tokenization, normalization, stemming, lemmatization
- Sequence-to-sequence models, text classification, named entity recognition, summarization
- Semantic search, embeddings, similarity metrics (cosine, Euclidean)
- Evaluation: BLEU, ROUGE, F1, perplexity

5. Distributed Systems & ML Infrastructure (for technical roles)

- Basics of distributed compute (data parallelism, model parallelism)
- Experience with GPUs/accelerators, HPC clusters, and Kubernetes for ML workloads
- Monitoring, scaling, and optimizing training/inference

6. Research, Experimentation, and Industry Awareness

- Reading and summarizing key AI/ML papers (start with “Attention is All You Need”)
- Fundamentals of setting up reproducible ML experiments and reporting metrics
- Awareness of major LLMs, model benchmarks, and responsible AI practices (privacy, security, fairness)
- Open-source ML contributions or collaborations (optional but highly valued)

7. End-to-End ML Product & Use Cases

- Pipelines: dataset curation, EDA, annotation, real-world data integration, and feedback loops
- API design and documentation, UX for AI-driven tools
- Specialization in enterprise applications: search, document automation, agent frameworks, chatbots

8. Soft Skills & Collaboration

- Communication: explaining complex ML topics to technical and non-technical audiences
- Collaboration: cross-functional teamwork (engineering, product, research, customer success)
- Project ownership, iterating through feedback, adapting to ambiguity
- Alignment with mission: interest in safe, scalable, enterprise-grade AI

How to Use This Roadmap

- Systematically learn & build projects for each topic above.
- For each technical segment, create 1–2 portfolio projects, e.g., "semantic search on custom data using RAG," "fine-tuned LLM for summarization," or "Kubernetes-powered model deployment."
- Document your code on GitHub, write clear READMEs, and if possible, contribute to open-source or participate in ML research competitions.
- Regularly read and summarize recent major LLM/NLP research (ArXiv, NeurIPS, ICLR), especially Cohere and competitor publications.
- Prepare to answer both technical and application-oriented interview questions as well as behavioral "mission alignment" questions.

If you tell me which team/role at Cohere you're most interested in, I can **prioritize the roadmap sections and recommend specific projects to feature on your resume.**

i am mostly interested in Member of Technical Staff, Software Engineer / Staff Software Engineer, AI Engineer / Applied ML Engineer, Product Designer / Technical Product Manager

Here's a tailored **topic and skill roadmap** to maximize your chances for the roles of **Member of Technical Staff, Software Engineer/Staff Software Engineer, AI Engineer/Applied ML Engineer, and Product Designer/Technical Product Manager at Cohere.**

1. Member of Technical Staff (MLE, Modeling, Data, Efficiency)

Key Skills & Topics

- Deep learning: transformers, LLMs, transfer learning, attention and embeddings
- Distributed ML: GPU computing, scaling, model/data parallelism, PyTorch/JAX
- Advanced NLP: semantic search, RAG, text generation, multilingual learning
- Model optimization: efficiency, quantization, pruning, benchmarking/compression
- Experimentation: designing reproducible ML experiments, benchmarking vs. state-of-the-art
- Ownership: end-to-end design, ML infrastructure, cross-team collaboration

Essential Projects

- Build and deploy a custom retrieval-augmented QA system on a business dataset
- Optimize a transformer for low-latency and low-resource inferencing (quantization + benchmarking)
- Design a distributed training pipeline for LLM finetuning on GPUs/kubernetes

2. Software Engineer / Staff Software Engineer (Infra, Platform, Applied ML)

Key Skills & Topics

- Strong programming fundamentals: Python, C++/Go, backend architecture, OOP, testing
- Distributed systems: cloud infra (AWS, GCP, Azure), Kubernetes container orchestration
- ML engineering: data pipelines, continuous integration, automation, CI/CD for ML
- API design: RESTful APIs, scalable microservices, secure authentication schemes
- Systems monitoring: logging, performance profiling, debugging production ML

Essential Projects

- Build a scalable ML pipeline (data ingestion, training, feedback) with cloud/Kubernetes deployment
- Design a backend API for hosting and serving a generative AI model with user authentication
- Implement real-time monitoring and scaling for inference workloads in a simulated production environment

3. AI Engineer / Applied ML Engineer

Key Skills & Topics

- NLP deployment: semantic search, text summarization, document QA, fine-tuning LLMs
- Productization: integrating LLMs into apps (SaaS, enterprise, B2B workflows)

- Evaluation: performance metrics, user-centric validation, robust testing
- Collaboration: syncing with product, engineering and client teams on features
- Emerging AI tools: open-source LLM frameworks, conversational agent toolkits

Essential Projects

- Deploy a semantic search application using an open-source LLM and custom corpus, with user-facing API
- Develop a chat summarization or FAQ generator feature using instruction-tuned LLMs
- Integrate an agent-based workflow for document review automation in a hypothetical business setting

4. Product Designer / Technical Product Manager

Key Skills & Topics

- ML-driven product management: user research, writing requirements, cross-functional leadership
- Design thinking: prototyping, UI/UX for AI tools, usability testing
- Technical fluency: understanding tradeoffs in ML products, system design basics
- Enterprise focus: mapping real business needs to AI-enabled product features
- Roadmapping: defining sprints, prioritizing features, managing releases

Essential Projects

- Lead the design and launch of the user interface for an LLM-powered search or agent product (Figma + basic frontend prototyping)
- Write a product spec and manage rollout for an automation feature using NLP in an enterprise context
- Conduct stakeholder/user research and translate needs into feature requirements/R&D priorities

Next Steps

- Choose 1–2 projects per role above to build a public portfolio (GitHub, blog, demo, Figma prototype)
- Document your process, results, and impact for each project—showcase both technical depth and product understanding
- Read Cohere's major tech blogs, recent ML papers, and relevant enterprise use cases, then try to reproduce or adapt one in your work
- Network with current Cohere engineers/designers, study their open-source or real-world contributions, and model your work accordingly

If you share your current technical strengths or backgrounds, I can customize this roadmap even further!

give for solutions architect as well

Here's a tailored roadmap for preparing for the **Solutions Architect** role at Cohere, along with recommended projects and topics.

Solutions Architect at Cohere: Roadmap & Key Topics

1. Technical ML/NLP Fundamentals

- Understanding LLM architectures and retrieval-augmented generation (RAG)
- NLP applications: semantic search, document automation, conversation agents
- ML pipeline design: data ingestion, preprocessing, evaluation and monitoring
- Cloud compute: deploying models on AWS/GCP/Azure, managing security and scaling

2. Enterprise Integration & Architecture

- Solution design: mapping customer business needs to Cohere APIs and LLM features
- Architectural patterns: microservices, API gateways, distributed data storage
- Security and compliance: data privacy, access control, enterprise onboarding

3. Client-Facing & Project Management Skills

- Leading technical workshops, demos, and proofs-of-concept (POCs)
- Writing solution architecture docs and technical proposals/requirements
- Communication: translating technical concepts for non-technical stakeholders

4. Collaboration & Cross-Functional Experience

- Coordination with engineering, product, and support teams for deployment
- Managing projects from needs assessment through integration and ongoing support
- Handling support/feedback loops and iteration with client engineering teams

Essential Projects for Your Portfolio

- **Enterprise NLP Pilot:** Architect and deploy a retrieval-augmented QA system for a simulated enterprise (e.g., internal company search or support bot). Document all integration steps and security measures.
- **End-to-End Model Deployment:** Lead an AI workflow migration to production in a cloud environment (AWS/GCP), including secure API exposure, scaling, and monitoring.
- **Technical Blueprint:** Produce a technical design document for implementing an LLM-powered agent, mapping business requirements to architecture and integration steps.

How to Use This Roadmap

- Build projects that demonstrate both technical depth (NLP/ML/Cloud integration) and client-facing architecture skills.
- Focus on presenting use-case-driven solutions, with detailed write-ups, diagrams, and results.
- Practice explaining architecture choices in non-technical terms—show you can bridge business and engineering.

If you need sample solutions architect resumes, technical blueprint samples, or want recommended online resources, let me know!

as a masters in Applied machine learning engineer and bachelors in Computer science and enginerrinng, graudting in may 2026, which role should i focus on

With a **Master's in Applied Machine Learning** (graduating May 2026), plus a **Bachelor's in Computer Science and Engineering**, you're **highly competitive for several roles at Cohere**. Here's a breakdown, focusing on strengths and strategy:

1. AI Engineer / Applied ML Engineer

Best-fit role for you—leverages both degrees

- Directly applies ML knowledge: building, fine-tuning, deploying LLMs and other models in real products.
- You'll work with production code, enterprise use cases, and be hands-on with APIs, data pipelines, and ML infrastructure.
- Portfolio: Focus on NLP projects, retrieval, chatbots, semantic search, and integration work.

2. Member of Technical Staff (MLE)

- If you have strong research, coding, and infrastructure experience—especially with transformers, distributed training, or large-scale ML—you're eligible.
- This is both hands-on engineering and the "brains" behind scaling SOTA ML solutions.
- Portfolio: Build ambitious ML systems, RAG, optimization, benchmarking, and demonstrate owning projects end-to-end.

3. Software Engineer (ML/Infra/GPT)

- If you like building and scaling ML platforms or infrastructure, especially with strong backend expertise (Python, cloud, distributed systems).
- Your CSE background gives you an edge here; supplement with ML/AI integrations or DevOps automation experience.

4. Solutions Architect (technical route)

- If you enjoy applied NLP/ML and want to **bridge technical and business worlds**—designing solutions, onboarding, client integration.
- This works well if you have experience translating technical features to customer value, or enjoy leadership/project management.

Product Designer / Technical Product Manager

- If you have a passion for product ideation, research, and want to drive enterprise AI strategy, these roles are open.
- You'll need user research/project management focus; if you enjoy solving "what/why/how" for enterprise LLM product decisions, consider this.

Recommendation: Focus on AI Engineer / Applied ML Engineer and Member of Technical Staff

- These roles maximize your ML and CS skillset, let you solve hard technical problems, and provide a direct path for new grads—especially those with advanced degrees and strong portfolios.
- You can pivot to Solutions Architect or Product roles later, after gaining production experience and enterprise exposure.

Tip:
Start applying as early as possible, build 2–3 high-impact projects around NLP/LLMs, and showcase contributions/results on GitHub. Also, consider research papers or open-source as bonus proof.

If you share specific technical areas you excel in (NLP, distributed systems, multi-modal, product, etc.), I can further tailor your application roadmap and project ideas!

then what all should i be proficient in for these roles, can you list them like a roadmap

Absolutely! Here's a detailed **roadmap of skills and proficiencies** you should develop to maximize your chances as an **AI Engineer / Applied ML Engineer** and **Member of Technical Staff (MLE)** at Cohere, with relevant additions for Software Engineer and Solutions Architect tracks.

Core Roadmap for Cohere Technical Roles

1. Essential Programming & CS Foundations

- Master Python (core language for ML), plus experience with C++, Java, or Go for systems work
- Strong grasp of data structures and algorithms, including complexity analysis
- Proficient in software engineering: OOP, design patterns, testing, debugging
- Git/version control; collaborative development, code reviews

2. Deep Learning & Machine Learning

- Strong working knowledge of ML fundamentals: supervised, unsupervised, and reinforcement learning
- Expertise with deep learning frameworks (PyTorch, TensorFlow, JAX)
- Ability to build, fine-tune, and deploy neural networks (MLPs, CNNs, RNNs, transformers)
- Practical experience with:
  - Training and evaluating LLMs (large language models, instruction tuning)
  - Retrieval-augmented generation (RAG) and grounding LLMs on external data
  - Model optimization: efficiency, quantization, benchmarking, scaling for inference

3. Natural Language Processing (NLP) & Applied ML

- Text preprocessing: tokenization, normalization, stemming, lemmatization
- Familiarity with open-source NLP toolkits (spaCy, HuggingFace)
- Semantic search, information retrieval, document/question answering, summarization
- Experience integrating NLP models into real-world applications (APIs, SaaS apps, enterprise plugins)

4. Machine Learning Infrastructure & Distributed Systems

- Building scalable ML pipelines (data ingestion, cleaning, automation, CI/CD)
- Familiarity with cloud platforms (AWS, GCP, Azure) and container orchestration (Docker, Kubernetes)
- Distributed training concepts: parallelism, cluster management, resource allocation
- Monitoring, profiling, and debugging ML in production (logging, metrics, alerts)

5. Research, Experimentation, & Metrics

- Design reproducible ML experiments; analyze results critically
- Evaluate models with proper metrics (accuracy, F1, BLEU, perplexity, latency/cost)
- Stay current with key ML papers ("Attention is All You Need", latest LLM developments), be able to replicate/adapt new research
- Open-source contributions or competitive ML participation (Kaggle, GitHub projects)

6. Product & Solution Awareness (plus for Solutions Architect)

- Ability to map technical solutions to enterprise needs and business impact
- Technical documentation: writing design specs, architectural diagrams, tutorials
- Client-facing skills: requirements gathering, demos, onboarding, technical support (especially for Solutions Architect)

7. Collaboration & Communication

- Communicate technical ideas to both engineers and business/product teams
- Experience with agile project management, iterative development, incorporating user feedback
- Cross-functional teamwork: collaborate with product managers, designers, customers

How to Use This Roadmap

- Build portfolio projects that clearly demonstrate each skill area (start with LLM chatbots, semantic search, scalable ML APIs)
- Document your process and results—publish on GitHub with clean READMEs and project walkthroughs
- Practice technical interviews and system design presentations
- Engage with research via reading, reproduction, and possibly publication

Pro-Tip:
For Member of Technical Staff and AI Engineer, emphasize **end-to-end ownership**: ideate,

build, test, deploy, and measure impact, not just model training. If you need specific sample projects for each topic, let me know!

1. https://cohere.com/about
2. https://en.wikipedia.org/wiki/Cohere
3. https://research.contrary.com/company/cohere
4. https://intuitionlabs.ai/articles/cohere-enterprise-ai-llm-profile
5. https://www.oracle.com/artificial-intelligence/cohere-aidan-gomez-interview/
6. https://www.entrepreneurship.artsci.utoronto.ca/news/how-cohere-ai-startup-co-founded-alumni-landed-170m-funding-year
7. https://exa.ai/websets/directory/cohere-funding
8. https://www.ia40.com/blog-podcast/cohere-co-founder-ivan-zhang
9. https://www.linkedin.com/company/cohere-ai/jobs
10. https://www.coherehealth.com/our-story
11. https://www.linkedin.com/jobs/view/member-of-technical-staff-mle-us-can-at-cohere-4246704563
12. https://www.linkedin.com/jobs/view/member-of-technical-staff-model-efficiency-at-cohere-4321671455
13. https://jobs.ashbyhq.com/cohere/d5cc437c-2397-4c1a-9d12-3a2bf64d98ea
14. https://www.linkedin.com/jobs/view/senior-member-of-technical-staff-multimodal-ai-at-cohere-4088517379
15. https://www.linkedin.com/jobs/view/member-of-technical-staff-modeling-at-cohere-4272083239
16. https://www.builtinsf.com/company/cohere-ai/jobs
17. https://www.builtinnyc.com/job/research-scholar/6849859
18. https://www.linkedin.com/jobs/view/senior-research-scientist-cohere-labs-at-cohere-4303837589
19. https://startup.jobs/company/cohere-2
20. https://www.remoterocketship.com/company/cohere-2/
21. https://cohere.com/careers
22. https://www.ziprecruiter.com/Jobs/Cohere-Ai
23. https://www.linkedin.com/jobs/view/director-community-engagement-at-cohere-4334706693
24. https://www.linkedin.com/company/cohere-ai/jobs
25. https://www.interviewquery.com/interview-guides/coherehealth-machine-learning-engineer
26. https://www.indeed.com/q-remote-cohere-jobs.html
27. https://www.linkedin.com/company/coherehealth/jobs
28. https://www.linkjob.ai/interview-questions/cohere-interview-process-and-questions/
29. https://www.linkedin.com/posts/jonathan-talmi-1a099852_cohere-member-of-technical-staff-data-activity-7087567471372603392-05Jb